

Software Engineering and Data Science
SEIS 766: Vision AI
Assignment #5

Due Date: March 19th

The dataset (**CellDNA.csv**) contains various measurements (i.e. size, center, etc) from thousands of bacterium under microscope. Write a program using either Python+Keras or MatLab to perform the **two-class** classification on this dataset using both **regular neural network** and **autoencoder**.

1. The **non-zero** values in the **last column** of "CellDNA.csv" are the target responses that indicate the bacterium (rows) that are interesting enough for further study. The 0s in the last column indicate the bacterium (rows) are NOT interesting candidates for further study. Convert this target dependent variable (last column) to **binary values** of either 0s or 1s for your **two-class** classification.
2. Use the 30 row indices in "**CellDNA_30rows_Indices.csv**" to select 30 rows of data from "CellDNA.csv". Do **NOT** split this 30 rows into training and testing since there are not enough records. Use **ALL** the 30 rows for training and testing.
3. Build your regular neural network and autoencoder with 4 layers. The number of neurons at each layer should be 10, 7, 4, and 2. Note, for autoencoder, it means that you have to build 4 neural networks and stack them up.
4. Use the same activation function of your choice either as sigmoid or relu in both your regular neural network and 4 autoencoders. But, for each AE, please carefully select an appropriate activation function at its output layer.
5. Train your regular neural network for 200 epochs (or 50 for optional experiment). Train each of your 4 autoencoder for 50 epochs. You can fine-tune your stacked autoencoder for another 50 epochs.
6. Use the same batch size and set learning rate to 10^{-3} for both NN and AE training.
7. Show the confusion matrices generated from both your regular neural network and autoencoder.

Submission Guideline:

1. Please include the answers to the above questions in a WORD document. Please also put **your name** on the top of your WORD document.
2. Please print your program (matlab or python) as **PDF / html** and include the **PDF / html** in your submission. Please name your program as "a5.m/.mlx/.py/.inpyb".
3. Please also include your program in the formats like .m/.mlx/.py/.inpyb in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.